

NOTE 03 — CIA TRIAD

What is the CIA Triad?

The CIA Triad is one of the fundamental concepts of information security.

CIA stands for:

- Confidentiality
- Integrity
- Availability

1. Confidentiality

Confidentiality means that information should only be accessible to authorized people or systems.

Examples

- Password protection
- Encryption
- Access control
- Multi-factor authentication
- File permissions

Example attack

An attacker steals a company's customer database.

This violates confidentiality because unauthorized people obtained the information.

2. Integrity

Integrity means that information remains accurate, complete, and protected from unauthorized modification.

Examples

- File integrity monitoring
- Hashes
- Digital signatures
- Database access controls
- Change management

Example attack

An attacker changes:

Salary = ₹50,000

to:

Salary = ₹500,000

This is an integrity violation.

3. Availability

Availability means authorized users can access systems and information when required.

Examples

- Backups
- Redundant systems
- Disaster recovery
- Load balancing
- DDoS protection
- High-availability infrastructure

Example attack

A DDoS attack overwhelms a web server so legitimate users cannot access the website.

This violates availability.

CIA Triad Table

Principle	Meaning	Example
Confidentiality	Prevent unauthorized access	Encryption
Integrity	Prevent unauthorized modification	File hash
Availability	Keep services accessible	Backup/redundancy

Real-world example

Consider an online banking application.

Confidentiality

Only the account owner should see account information.

Integrity

A user should not be able to modify their account balance without authorization.

Availability

The banking application should remain accessible when customers need it.

Conclusion

The CIA Triad provides a basic framework for thinking about security requirements and security incidents.